

Project Management Business Requirements Document (BRD)

DOCUMENT CONTROL

1.1. Change Record

Date	Author	Version	Change Reference

1.2. Reviewers

Name	Position

1. Introduction and background

This Business Requirements Document (BRD) outlines the business requirements for the Project Module within the Real Estate Project Management System (RE-PMS), which will be developed for **WUJHA DEVELOPMENT**.

The Project Module is designed to streamline and optimize critical aspects of real estate project management, including planning, scheduling, budgeting, resource management, risk management, and collaboration. It will serve as a centralized platform that enables project teams to efficiently initiate, plan, execute, monitor, and close real estate development projects.

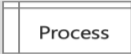
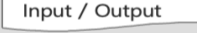
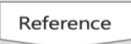
By implementing this module, **WUJHA DEVELOPMENT** aims to achieve greater visibility and control over project activities, ensuring projects are completed on time, within budget, and according to quality standards. The system will facilitate better decision-making through real-time data insights, reduce manual processes, enhance collaboration among stakeholders, and minimize risks associated with project delays or budget overruns.

Ultimately, the Project Module will support **WUJHA DEVELOPMENT** in delivering successful real estate projects with improved efficiency, accountability, and transparency.

Document Process Flow Diagram

Process flow diagram supports the analysis of the TO-BE business process by:

- Clearly communicating the current business processes
- Allow analysts to document the current business in a structured way
- Facilitate business requirements mapping to the application

Process Symbol	Definition
	This symbol starts the process.
	This symbol represents a group of steps or activities to perform work (i.e. a process).
	This symbol represents an activity to perform work defined in detail. This symbol may represent automated or manual activity.
	This symbol represents a decision . Decisions are typically phased as “yes/no” questions. Look for a “yes” path and a “no” path out of this symbol
	This symbol represents information input / output , such as a report or document. This symbol does not always imply hard copy, as some are verbal.
	This symbol represents an off page reference. Indicates a cross-reference with another page of the flowchart.
	This represents an input or output symbol of “ Information Entered into System ” or “ Information Received from System ”.
	This symbol concludes the process .

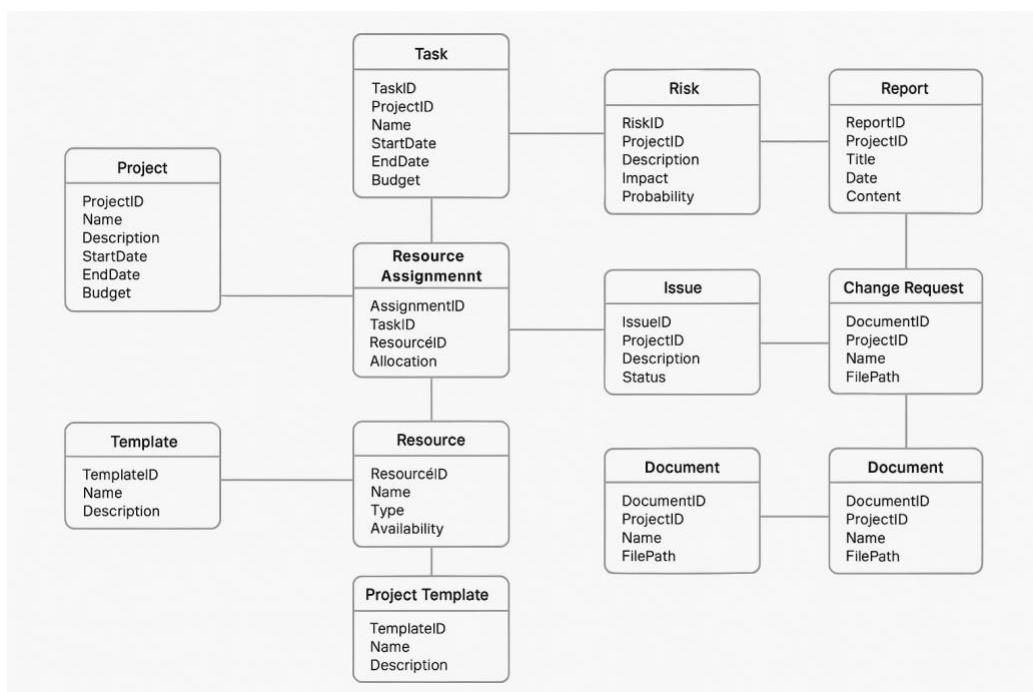
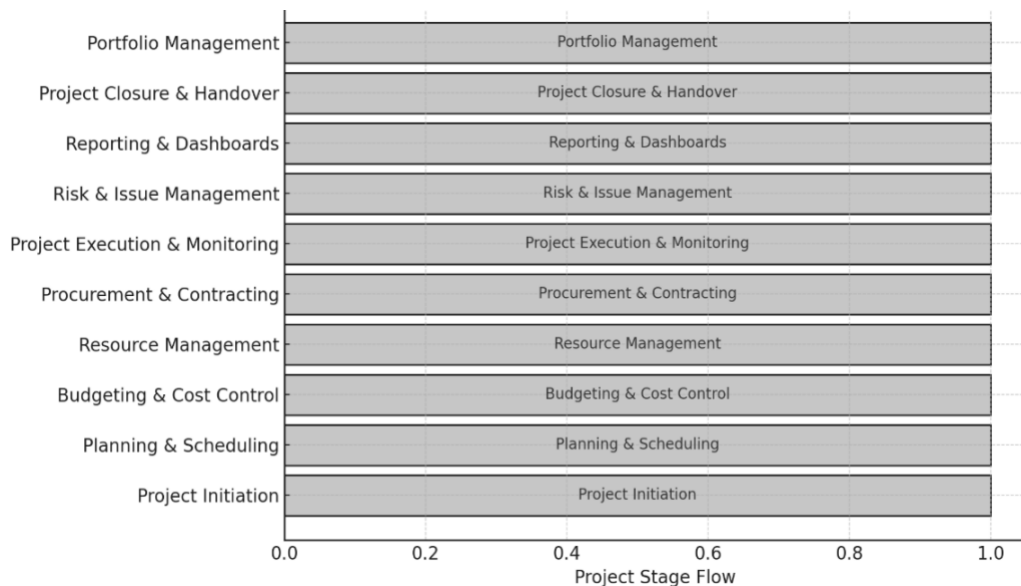
1. Purpose

This document outlines the **business requirements** and **to-be processes** for the PM solution implementation at Wujha. The focus is on the **future state** of projects operations after system implementation, detailing automated workflows and new, streamlined processes aimed at improving PM functions.

Business Objectives:

- 1.1. **Improve Project Planning and Scheduling:** Enable project managers to create detailed project plans, define milestones, allocate tasks, and establish timelines effectively. The system will provide visual tools such as Gantt charts and project calendars to facilitate accurate planning and timely execution.
- 1.2. **Enhance Budget Control and Cost Management:** Implement budget tracking and cost management functionalities to monitor project expenses against predefined budgets. Real-time financial dashboards and alerts for budget deviations will help in controlling costs and maximizing profitability.
- 1.3. **Optimize Resource Management:** Provide a centralized resource management system to allocate human resources, equipment, and materials efficiently across multiple projects. The module will enable tracking of resource utilization, availability, and workload balancing to improve productivity.
- 1.4. **Strengthen Risk Management:** Incorporate a risk management framework within the system to identify, assess, and mitigate potential risks early in the project lifecycle. Risk registers, automated alerts, and contingency planning tools will help minimize project disruptions.
- 1.5. **Increase Collaboration and Communication:** Facilitate seamless collaboration among project teams, stakeholders, contractors, and consultants through shared workspaces, document management, and communication tools. Enhanced collaboration will ensure all stakeholders remain informed and aligned.
- 1.6. **Provide Real-Time Reporting and Analytics:** Deliver real-time reporting, dashboards, and analytics capabilities to provide insights into project performance, resource utilization, financial status, and risk exposure. Data-driven insights will support faster and more informed decision-making.
- 1.7. **Ensure Scalability and Flexibility:** Design the Project Module to be scalable and flexible, allowing WUJHA DEVELOPMENT to easily adapt the system to different types and sizes of real estate projects as the organization grows or project complexity increases.
- 1.8. **Improve Compliance and Documentation:** Ensure that all project activities are properly documented and compliant with industry standards, regulations, and internal policies. The system will offer audit trails, version control, and automated document templates to support compliance efforts.
- 1.9. **Scope of Work:** This module will cover the following functionalities:
 - **Project Planning & Scheduling:** Work Breakdown Structure (WBS), activity definition, dependency management, critical path scheduling, and baseline tracking.
 - **Resource Management:** Allocation, leveling, and productivity tracking for human, material, and equipment resources.
 - **Cost Management & Budgeting:** Cost estimation, Earned Value Management (EVM), and budget tracking with integration to financial systems.
 - **Risk Management:** Risk identification, assessment, and mitigation strategies using Monte Carlo simulations.
 - **Collaboration & Communication:** Multi-user access, document management, and workflow automation.
 - **Reporting & Dashboards:** Pre-built and custom reports, executive dashboards, and real-time project monitoring.
 - **Stakeholders**
 - **Project Managers:** Oversee project planning, scheduling, and execution.
 - **Developers & Contractors:** Collaborate on task execution and resource allocation.
 - **Financial Teams:** Track and manage budgets, costs, and forecasts.
 - **Executives & Stakeholders:** Monitor project progress through real-time reporting and dashboards.
 - **Regulatory Authorities:** Ensure compliance with laws and regulations.

1.10. High-Level Project Process Activities Flow and Steps:



1. Detailed Descriptions of Projects Module Functionalities

Functionality	Description for Development
EPS (Enterprise Project Structure)	Hierarchical breakdown of the entire portfolio of projects across the organization. Allows grouping projects by division, geography, client, or type for consolidated tracking and reporting. Enables user access controls at each level.
Portfolio Management	Capability to manage multiple projects under a single umbrella. Supports prioritization, monitoring project health across portfolio, risk exposure analysis, and resource balancing at portfolio level. Includes scenario planning for better decision-making.
WBS (Work Breakdown Structure)	Decomposes project deliverables into manageable work packages and activities. Provides clear visibility into scope hierarchy, dependencies, responsibilities, and timelines. Essential for scheduling and cost allocation.
Project Structuring	Framework to set up new projects including project codes, templates, WBS, calendars, resource allocation framework, and predefined workflows. Supports standardized setup for consistency and control.
Critical Path Method (CPM)	Scheduling algorithm that identifies the longest sequence of dependent activities and calculates the earliest possible completion date. Helps highlight critical tasks and potential delays.
Baseline Management	Captures an initial approved plan (scope, time, cost) for comparison against actual progress and performance. Supports multiple baselines for scope changes and re-planning.
Resource Assignments	Assigns labor, equipment, and materials to specific activities. Tracks availability, utilization rates, costs, and efficiency of resource deployment.
Resource Pool Management	Centralized resource repository. Tracks competencies, availability, locations, and costs. Supports cross-project resource sharing and optimal allocation.
Resource Leveling	Automatic adjustment of resource assignments to resolve over-allocations and optimize schedules without manual rework.
Resource Usage Profiles	Graphical visualization of resource allocation and consumption over time. Aids in spotting peaks, underuse, or overloads early.
Procurement Management	Tracks procurement items, RFQs, purchase orders, supplier details, and delivery schedules. Links procurement directly to WBS activities and cash flow requirements.
Contract Management	Manages supplier and subcontractor contracts, including terms, milestones, payments, and variations. Provides alerts for contract deadlines and compliance.
Supplier Database Integration	Maintains supplier records with performance history, qualifications, certifications, and contact information. Enables vendor selection and evaluation automation.
Cost Loading to Schedule	Assigns project costs directly to activities in the schedule. Enables precise cash flow projections and earned value calculations.
Time-Phased Budgeting	Spreads costs and resource demands over time based on project timelines. Supports cash flow forecasting and financial planning.
Earned Value Management (EVM)	Integrated performance measurement system that compares planned progress and costs to actual performance. Delivers metrics like CPI (Cost Performance Index) and SPI (Schedule Performance Index).
Progress Updating	Real-time input of project status updates, including actual start/finish dates, remaining duration, and physical % complete for activities.
Field Progress Tracking	Mobile-enabled updates from the construction site for daily logs, inspections, quality, and safety checks. Integrates directly with project schedule and budget.
Impact Analysis	Simulation of the impact of changes (delays, scope adjustments, cost increases) on project baseline, critical path, and overall objectives.
Risk Register	Central log of identified risks with probability, impact ratings, owners, mitigation actions, and status. Enables automated risk reporting and escalation.
Mitigation Plans	Predefined response strategies for high-probability risks, tracked for implementation and effectiveness. Integrated with workflow automation.
Customizable Dashboards	Visual, real-time presentation of project KPIs, risks, milestones, and financial data. Configurable per user role (PM, Executive, Finance, etc.).
Pre-configured Reports	Standardized reporting templates for schedules, costs, risks, procurement, resources, and earned value performance. Supports both detail and executive summaries.
Portfolio Reporting	Consolidated reports across multiple projects showing overall health, status, risks, and opportunities. Enables strategic decision-making at program level.
Closure Checklists	Structured list of tasks to formally close the project, including contractual, financial, technical, and administrative steps.

1.1. Process steps based on the Projects functionalities:

Functionality	Process Steps	Stages
1. Project Structuring	1. Create Enterprise Project Structure (EPS) to categorize projects.	Project Initialization: Setup EPS and WBS.
	2. Define and organize projects into Portfolio, Program, and Project levels.	Configuration: Align project structure and management hierarchies.
2. Portfolio Management	1. Consolidate project data across portfolios.	Portfolio Creation: Assign projects to portfolios.
	2. Monitor portfolio performance against key metrics.	Portfolio Monitoring: Track progress, issues, and performance.
3. WBS Creation and Cost Loading	1. Define Work Breakdown Structure (WBS) with task dependencies.	WBS Design: Create detailed work breakdown.
	2. Assign resources and costs to WBS elements.	Cost Estimation & Allocation: Allocate budgets to WBS elements.
4. Scheduling and CPM	1. Set project schedule with start and end dates for tasks.	Schedule Planning: Define and structure tasks with dependencies.
	2. Calculate Critical Path using CPM to identify critical tasks.	Critical Path Calculation: Identify critical path tasks and dependencies.
5. Resource Assignment & Pool Management	1. Define resource pools (personnel, equipment, etc.).	Resource Planning: Assign roles to tasks.
	2. Allocate resources to specific tasks.	Resource Optimization: Analyze and adjust resource allocation.
6. Cost Control (EVM)	1. Link costs to schedule tasks.	Cost Tracking: Link financial data to project timelines.
	2. Use Earned Value Management (EVM) to track performance.	Performance Monitoring: Use EVM to track variances and performance.
7. Procurement & Contract Management	1. Create procurement plans and define requirements for suppliers.	Procurement Setup: Define procurement needs and vendor selection.
	2. Set up contracts for suppliers and vendors.	Contract Management: Manage and track supplier relationships.
8. Progress Updating & Field Tracking	1. Regularly update project progress (daily/weekly).	Progress Reporting: Document task completion and milestones.
	2. Track field progress via reports or mobile apps.	Field Data Collection: Gather updates from project managers on-site.
9. Risk Management	1. Maintain a Risk Register to document potential risks.	Risk Identification: Identify and assess risks.
	2. Perform impact analysis to evaluate the severity and likelihood of risks.	Mitigation Planning: Develop plans to reduce or avoid risks.
10. Reporting & Dashboards	1. Create customizable dashboards for project monitoring.	Dashboard Setup: Define KPIs for tracking.
	2. Use pre-configured reports to track project health and performance.	Reporting: Generate and distribute periodic reports to stakeholders.
11. Project Closure & Lessons Learned	1. Complete closure checklists to ensure all project elements are done.	Closure: Ensure all contractual and project tasks are completed.
	2. Document lessons learned and best practices for future projects.	Knowledge Transfer: Store lessons learned for future use.
12. Scenario Planning & Benchmarking	1. Create scenario plans to evaluate different project outcomes.	Scenario Planning: Define various "what-if" scenarios.
	2. Benchmark project performance using industry standards or best practices.	Benchmarking: Compare project performance against best practices or competitors.

1.2. Visual Data Flow Process with the assigned Roles and Automation

Process Stage	Action	Role(s) Responsible	Automation Touchpoint	Data Flow Output
1. Project Initialization	Setup EPS and WBS	Project Manager, Admin	Auto-generate EPS/WBS templates	Configured EPS/WBS structure
	Define Project Structure	Project Manager, Portfolio Manager	Auto-align with organizational structure	Organized structure based on hierarchy
	Create Project Metadata	Project Manager, Admin	Auto-create Project ID and metadata	Project data populated
2. Portfolio Management	Assign Project to Portfolio	Portfolio Manager, Project Manager	Auto-assign project to relevant portfolio	Portfolio assigned, visible in Portfolio dashboard
	Monitor Portfolio Progress	Portfolio Manager	Real-time dashboard update	Portfolio progress data
	Review Portfolio Health	Portfolio Manager	Auto-generate health metrics	Portfolio health report
3. Planning & Estimation	Design WBS	Project Manager, Planning Engineer	Auto-suggest WBS elements based on templates	WBS structured, with task dependencies
	Estimate Project Cost	Financial Analyst, Cost Estimator	Auto-calculate cost estimates	Estimated cost based on WBS
	Develop Project Schedule	Scheduler, Project Manager	Auto-schedule tasks, based on dependencies	Generated project schedule
	Calculate Critical Path	Project Scheduler, Planner	Auto-identify critical path	Critical path data and alert
4. Resource Management	Assign Resources to Tasks	Resource Manager, Project Manager	Auto-assign resources to tasks based on capacity	Resource allocation and availability
	Optimize Resource Utilization	Resource Manager, Planner	Resource optimization suggestions	Resource utilization report
	Validate Resource Availability	Resource Manager	Auto-check resource availability	Resource availability report
5. Cost & Performance Management	Track Actual vs. Baseline Cost	Financial Analyst, Project Manager	Auto-link actual vs. baseline cost	Cost tracking report
	Calculate EVM (Earned Value Management)	Financial Analyst, Project Manager	Auto-calculate SPI/CPI metrics	SPI, CPI metrics generated
6. Procurement & Contracting	Setup Procurement Plan	Procurement Manager, Project Manager	Auto-match suppliers to requirements	Procurement plan, suppliers matched
	Select Vendors and Suppliers	Procurement Manager, Admin	Auto-suggest vendors based on historical data	Vendor list and selection
	Manage Contracts	Procurement Manager, Legal	Contract management automation	Supplier contracts, signed documents
7. Execution & Monitoring	Update Task Completion Percentage	Project Manager, Field Manager	Auto-update progress based on field data	Updated task progress data
	Collect Field Data	Field Manager, Project Manager	Auto-sync field data with project database	Field progress data
	Monitor KPIs and Adjustments	Project Manager, Performance Analyst	Real-time KPI updates	KPI dashboard data
8. Risk Management	Identify & Assess Risks	Risk Manager, Project Manager	Auto-assign risk severity scores	Risk register, risk heatmap
	Develop Mitigation Plan	Risk Manager, Project Manager	Auto-suggest mitigation strategies	Risk mitigation plan
	Assess Risk Impact	Risk Manager, Project Manager	Auto-calculate risk impact	Risk impact analysis

9. Reporting & Analytics	Generate Reports	Project Manager, Reporting Analyst	Auto-generate reports based on predefined criteria	Project status reports, financial reports
	Update Dashboards	Reporting Analyst, Project Manager	Auto-refresh dashboard with real-time data	Real-time updated dashboard
	Analyze "What-if" Scenarios	Project Manager, Performance Analyst	Auto-run scenario analysis models	Scenario results
10. Closure & Knowledge Management	Complete Project Closure Checklist	Project Manager, Admin	Auto-generate closure checklist	Closure tasks checklist
	Archive Project Documents	Project Manager, Admin	Auto-archive project documents	Archived project records
	Document Lessons Learned	Project Manager, Knowledge Manager	Auto-collect lessons learned from project logs	Lessons learned repository

1.3. Step-by-step breakdown for each stage:

Stage	Detailed Process Steps
Project Initialization: Setup EPS and WBS	1. Define the overall project goals and objectives. 2. Establish Enterprise Project Structure (EPS). 3. Create initial Work Breakdown Structure (WBS) at high level. 4. Assign project teams to EPS elements.
Configuration: Align project structure and management hierarchies	1. Define project roles and responsibilities. 2. Establish approval hierarchies and workflows. 3. Configure project governance structure. 4. Map organizational units to EPS.
Portfolio Creation: Assign projects to portfolios	1. Group projects based on strategy, geography, or function. 2. Assign projects to respective portfolios. 3. Define portfolio-level objectives and success criteria.
Portfolio Monitoring: Track progress, issues, and performance	1. Set portfolio-level KPIs. 2. Monitor project health across the portfolio. 3. Identify underperforming projects. 4. Initiate corrective actions or escalations.
WBS Design: Create detailed work breakdown	1. Decompose high-level WBS into detailed tasks and deliverables. 2. Assign ownership to each WBS element. 3. Validate WBS with project stakeholders.
Cost Estimation & Allocation: Allocate budgets to WBS elements	1. Estimate costs for each WBS component. 2. Allocate budgets accordingly. 3. Align cost estimates with resource and procurement plans. 4. Validate total project budget.
Schedule Planning: Define and structure tasks with dependencies	1. Sequence project activities. 2. Define task durations. 3. Set dependencies and constraints. 4. Develop the preliminary project schedule.
Critical Path Calculation: Identify critical path tasks and dependencies	1. Perform network analysis. 2. Identify critical tasks with zero float. 3. Highlight high-risk dependencies. 4. Review and validate critical path with team.
Resource Planning: Assign roles to tasks	1. Identify required resources (labor, equipment). 2. Assign roles and responsibilities to tasks. 3. Ensure availability of assigned resources.
Resource Optimization: Analyze and adjust resource allocation	1. Analyze resource usage across project. 2. Resolve overallocations or shortages. 3. Optimize assignments for efficiency. 4. Document resource adjustments.
Cost Tracking: Link financial data to project timelines	1. Integrate budget with project schedule. 2. Track actual spend versus planned. 3. Update financial records regularly. 4. Report cost variances.

Performance Monitoring: Use EVM to track variances and performance	1. Implement Earned Value Management (EVM). 2. Monitor planned vs. actual progress. 3. Calculate CPI and SPI indicators. 4. Report performance metrics.
Procurement Setup: Define procurement needs and vendor selection	1. Identify procurement requirements. 2. Prepare and issue RFPs/RFQs. 3. Evaluate vendor proposals. 4. Award contracts to selected vendors.
Contract Management: Manage and track supplier relationships	1. Establish contract terms and conditions. 2. Monitor contract compliance. 3. Manage supplier performance. 4. Resolve disputes and changes.
Progress Reporting: Document task completion and milestones	1. Define reporting frequency. 2. Collect task completion updates. 3. Update project dashboards. 4. Share progress reports with stakeholders.
Field Data Collection: Gather updates from project managers on-site	1. Set up field data capture tools. 2. Train field staff on data submission. 3. Collect daily or weekly progress data. 4. Integrate field data with project system.
Risk Identification: Identify and assess risks	1. Conduct risk workshops. 2. Document risks in Risk Register. 3. Classify risks by impact and likelihood. 4. Assign owners for risk monitoring.
Mitigation Planning: Develop plans to reduce or avoid risks	1. Develop risk response strategies. 2. Assign mitigation actions. 3. Monitor implementation of mitigation plans. 4. Update risk register regularly.
Dashboard Setup: Define KPIs for tracking	1. Define project and portfolio KPIs. 2. Design dashboard layout. 3. Configure data sources. 4. Test dashboard functionality.
Reporting: Generate and distribute periodic reports to stakeholders	1. Set reporting schedule. 2. Generate performance and financial reports. 3. Distribute reports to project teams and management. 4. Collect feedback for improvements.
Closure: Ensure all contractual and project tasks are completed	1. Verify completion of all deliverables. 2. Finalize contract closures. 3. Release project resources. 4. Conduct closure meeting with stakeholders.
Knowledge Transfer: Store lessons learned for future use	1. Conduct lessons learned workshops. 2. Document successes and challenges. 3. Store findings in knowledge repository. 4. Share with organizational stakeholders.
Scenario Planning: Define various "what-if" scenarios	1. Identify critical project variables. 2. Develop alternative scenarios. 3. Analyze impact of each scenario. 4. Prepare contingency plans.
Benchmarking: Compare project performance against best practices or competitors	1. Select benchmarking criteria. 2. Collect internal and external data. 3. Analyze gaps and opportunities. 4. Implement improvements based on findings.

1.4. List of stages and detailed process steps mapped with assigned roles

Stage	Detailed Process Steps	Assigned Role
Project Initialization: Setup EPS and WBS	1. Define the overall project goals and objectives	Project Manager, Project Sponsor
	2. Establish Enterprise Project Structure (EPS)	Project Manager, Portfolio Manager
	3. Create initial Work Breakdown Structure (WBS) at high level	Project Manager, Project Scheduler
	4. Assign project teams to EPS elements	HR Manager, Project Manager
Configuration: Align project structure and management hierarchies	1. Define project roles and responsibilities	HR Manager, Project Manager
	2. Establish approval hierarchies and workflows	Project Manager, Business Analyst
	3. Configure project governance structure	Project Manager, Governance Officer
	4. Map organizational units to EPS	Project Manager, Business Analyst
Portfolio Creation: Assign projects to portfolios	1. Group projects based on strategy, geography, or function	Portfolio Manager, Strategy Lead
	2. Assign projects to respective portfolios	Portfolio Manager
	3. Define portfolio-level objectives and success criteria	Portfolio Manager, Project Sponsor
Portfolio Monitoring: Track progress, issues, and performance	1. Set portfolio-level KPIs	Portfolio Manager
	2. Monitor project health across the portfolio	Portfolio Manager, Project Manager
	3. Identify underperforming projects	Portfolio Manager
	4. Initiate corrective actions or escalations	Portfolio Manager, Project Manager
WBS Design: Create detailed work breakdown	1. Decompose high-level WBS into detailed tasks and deliverables	Project Scheduler, Project Manager
	2. Assign ownership to each WBS element	Project Manager
	3. Validate WBS with project stakeholders	Project Manager, Stakeholder Representatives
Cost Estimation & Allocation: Allocate budgets to WBS elements	1. Estimate costs for each WBS component	Cost Estimator, Project Manager
	2. Allocate budgets accordingly	Finance Manager, Project Manager
	3. Align cost estimates with resource and procurement plans	Project Manager, Procurement Manager
	4. Validate total project budget	Finance Manager, Project Manager
Schedule Planning: Define and structure tasks with dependencies	1. Sequence project activities	Project Scheduler
	2. Define task durations	Project Scheduler
	3. Set dependencies and constraints	Project Scheduler
	4. Develop the preliminary project schedule	Project Scheduler, Project Manager
Critical Path Calculation: Identify critical path tasks and dependencies	1. Perform network analysis	Project Scheduler
	2. Identify critical tasks with zero float	Project Scheduler
	3. Highlight high-risk dependencies	Project Scheduler
	4. Review and validate critical path with team	Project Manager, Project Scheduler
Resource Planning: Assign roles to tasks	1. Identify required resources (labor, equipment)	Resource Manager
	2. Assign roles and responsibilities to tasks	Project Manager, Resource Manager
	3. Ensure availability of assigned resources	Resource Manager
Resource Optimization: Analyze and adjust resource allocation	1. Analyze resource usage across project	Resource Manager
	2. Resolve overallocations or shortages	Resource Manager
	3. Optimize assignments for efficiency	Resource Manager
	4. Document resource adjustments	Resource Manager
Cost Tracking: Link financial data to project timelines	1. Integrate budget with project schedule	Finance Manager, Project Manager
	2. Track actual spend versus planned	Finance Manager
	3. Update financial records regularly	Finance Manager
	4. Report cost variances	Project Manager, Finance Manager
Performance Monitoring: Use EVM to track variances and performance	1. Implement Earned Value Management (EVM)	Project Manager, Cost Control Specialist
	2. Monitor planned vs. actual progress	Project Manager
	3. Calculate CPI and SPI indicators	Cost Control Specialist
	4. Report performance metrics	Project Manager

Procurement Setup: Define procurement needs and vendor selection	1. Identify procurement requirements	Procurement Manager, Project Manager
	2. Prepare and issue RFPs/RFQs	Procurement Manager
	3. Evaluate vendor proposals	Procurement Manager, Project Manager
	4. Award contracts to selected vendors	Procurement Manager, Project Manager
Contract Management: Manage and track supplier relationships	1. Establish contract terms and conditions	Procurement Manager
	2. Monitor contract compliance	Procurement Manager
	3. Manage supplier performance	Procurement Manager
	4. Resolve disputes and changes	Procurement Manager, Project Manager
Progress Reporting: Document task completion and milestones	1. Define reporting frequency	Project Manager
	2. Collect task completion updates	Project Manager, Project Team
	3. Update project dashboards	Project Manager
	4. Share progress reports with stakeholders	Project Manager
Field Data Collection: Gather updates from project managers on-site	1. Set up field data capture tools	Project Manager, IT Support
	2. Train field staff on data submission	Project Manager, HR Manager
	3. Collect daily or weekly progress data	Site Manager, Field Staff
	4. Integrate field data with project system	IT Support, Project Manager
Risk Identification: Identify and assess risks	1. Conduct risk workshops	Risk Manager, Project Manager
	2. Document risks in Risk Register	Risk Manager
	3. Classify risks by impact and likelihood	Risk Manager
	4. Assign owners for risk monitoring	Risk Manager, Project Manager
Mitigation Planning: Develop plans to reduce or avoid risks	1. Develop risk response strategies	Risk Manager
	2. Assign mitigation actions	Risk Manager
	3. Monitor implementation of mitigation plans	Risk Manager, Project Manager
	4. Update risk register regularly	Risk Manager
Dashboard Setup: Define KPIs for tracking	1. Define project and portfolio KPIs	Portfolio Manager, Project Manager
	2. Design dashboard layout	Business Analyst, Project Manager
	3. Configure data sources	IT Support, Project Manager
	4. Test dashboard functionality	IT Support, Business Analyst
Reporting: Generate and distribute periodic reports to stakeholders	1. Set reporting schedule	Project Manager
	2. Generate performance and financial reports	Finance Manager, Project Manager
	3. Distribute reports to project teams and management	Project Manager
	4. Collect feedback for improvements	Project Manager
Closure: Ensure all contractual and project tasks are completed	1. Verify completion of all deliverables	Project Manager
	2. Finalize contract closures	Procurement Manager, Project Manager
	3. Release project resources	Project Manager, HR Manager
	4. Conduct closure meeting with stakeholders	Project Manager
Knowledge Transfer: Store lessons learned for future use	1. Conduct lessons learned workshops	Project Manager, Knowledge Manager
	2. Document successes and challenges	Project Manager, Knowledge Manager
	3. Store findings in knowledge repository	Knowledge Manager
	4. Share with organizational stakeholders	Knowledge Manager
Scenario Planning: Define various "what-if" scenarios	1. Identify critical project variables	Project Manager
	2. Develop alternative scenarios	Project Manager, Business Analyst
	3. Analyze impact of each scenario	Project Manager, Risk Manager
	4. Prepare contingency plans	Project Manager, Risk Manager
Benchmarking: Compare project performance against best practices or competitors	1. Select benchmarking criteria	Project Manager, Portfolio Manager
	2. Collect internal and external data	Business Analyst, Project Manager
	3. Analyze gaps and opportunities	Business Analyst, Project Manager
	4. Implement improvements based on findings	Project Manager, Portfolio Manager

1.5. Process Steps mapped with the Functionalities

#	Process Stage	Key Activities (Enhanced)	Projects Functionality Reference	Automation Opportunities
1	Project Initiation	- Capture project charter - Feasibility analysis - Generate project code & WBS - Assign PM & team - Define portfolio alignment	✓ Project Structuring ✓ EPS (Enterprise Project Structure) ✓ Portfolio Management	- Auto-code generation - Approval workflows - Stakeholder notifications
2	Planning & Scheduling	- Define work packages - Activity sequencing & dependencies - Set milestones & constraints - Resource assignments - Develop baseline schedule	✓ WBS Creation ✓ Critical Path Method (CPM) ✓ Baseline Management ✓ Resource Assignments	- Template-driven planning - Baseline auto-capture - Dependency alerts
3	Budgeting & Cost Control	- Define CBS (Cost Breakdown Structure) - Integrate time-phased budgeting - Establish budget baselines - Forecasting & cash flow analysis - Manage change orders	✓ Cost Loading to Schedule ✓ Time-Phased Budgeting ✓ Earned Value Management (EVM)	- Budget imports - Variance analysis reports - Alerts on overruns
4	Resource Management	- Define resource pool - Resource leveling - Track availability & usage - Equipment tracking - Monitor subcontractor performance	✓ Resource Pool Management ✓ Resource Leveling ✓ Resource Usage Profiles	- Resource dashboards - Availability alerts - Subcontractor performance tracking
5	Procurement & Contracting	- Prepare procurement plan - Supplier selection & evaluation - Issue RFQs / POs - Contract award & monitoring - Link contracts to activities	✓ Procurement Management ✓ Contract Management ✓ Supplier Database Integration	- Auto-PO generation - Contract alerts - Milestone-based payment triggers
6	Project Execution & Monitoring	- Update progress (physical & financial) - Actual vs. planned comparison - Conduct inspections - Field reporting (quality, safety) - Review critical path	✓ Progress Updating ✓ Critical Path Analysis ✓ Field Progress Tracking	- Mobile reporting - Auto-progress updates - Real-time dashboards
7	Risk & Issue Management	- Identify risks - Qualitative & quantitative risk analysis - Apply mitigation plans - Track issues & escalations	✓ Risk Register ✓ Impact Analysis ✓ Mitigation Plans	- Risk alerts - Auto-assignment of mitigation tasks - Issue escalation workflows
8	Reporting & Dashboards	- Standard and ad-hoc reports - Earned Value analysis - Project portfolio overview - Custom KPIs	✓ Customizable Dashboards ✓ Pre-configured Reports ✓ Portfolio Reporting	- Scheduled report generation - KPI monitoring and alerts
9	Project Closure & Handover	- Confirm scope completion - Final reconciliation (time, cost, quality) - Lessons learned - Document archival - Warranty & facility handover	✓ Closure Checklists ✓ Archive Management ✓ Lessons Learned Repository	- Auto-generated closure reports - Document archival automation - Warranty reminders
10	Portfolio Management <i>(Optional)</i>	- Monitor all projects in portfolio - Scenario planning (what-if analysis) - Benchmark performance against standards	✓ Portfolio Management ✓ Scenario Planning Tools ✓ Benchmarking Capabilities	- Auto-prioritization - Scenario analysis tools - Portfolio-level dashboards

1.6. Pre-requisite Information & Setup Needed

Step	Pre-requisite Information Needed
Project Initialization	- Organizational hierarchy (EPS) - Project charter / initial brief - Preliminary WBS template
Configuration	- Project governance structure - Roles & responsibilities matrix (RACI) - Approval workflows
Portfolio Creation	- List of active and planned projects - Portfolio categories (e.g., strategic, operational)
Portfolio Monitoring	- Defined KPIs - Project baselines (scope, time, cost)
WBS Design	- Scope statement - Deliverables list - Standard WBS dictionary
Cost Estimation & Allocation	- Historical cost data - Resource rates - Scope baseline
Schedule Planning	- Activity list - Milestones - Task durations & dependencies
Critical Path Calculation	- Finalized activity network - Task durations
Resource Planning	- Resource pool - Skills inventory - Availability calendar
Resource Optimization	- Resource allocation plan - Risk-adjusted schedules
Cost Tracking	- Financial system integration - Baseline budget
Performance Monitoring	- EVM setup (planned value, earned value, actual cost)
Procurement Setup	- Procurement strategy - Vendor database
Contract Management	- Signed contracts - Contract obligations tracker
Progress Reporting	- Reporting formats - Communication plan
Field Data Collection	- Data collection tools - Field reporting protocols
Risk Identification	- Risk register template - Historical risk data
Mitigation Planning	- Approved risk responses - Contingency reserves
Dashboard Setup	- Reporting tools access - Defined KPIs and data sources
Reporting	- Reporting calendar - Audience identification
Closure	- Closure checklist - Acceptance documentation
Knowledge Transfer	- Lessons learned template - Project archives
Scenario Planning	- Alternative strategies - Sensitivity analysis tools
Benchmarking	- Industry benchmarks - Competitive data

1.7. Automation Workflow— Project Management

#	Process Step	Trigger	Automation Action	Notification / Escalation	Integration / Output
1	Project Initialization	New Project Request approved	Auto-generate Project Code & ID; Create project folder	Notify PMO, PJM, Finance, Procurement	Document Management System,
2	Configuration	Project approved	Auto-generate templates (WBS, Budget, Timeline); Assign roles	Notify PJM	HR System
3	Portfolio Creation	Portfolio initiated	Auto-link projects; Create portfolio dashboard	Notify PMO, DIR	Dashboard system
4	WBS Design	WBS creation initiated	Template suggestion; Auto-number WBS; Approval workflow	Notify PMO for review	Project module
5	Cost Estimation & Allocation	WBS approved	Auto-populate cost template; Budget validation	Notify Finance	ERP System
6	Schedule Planning	Cost plan approved	Generate Gantt chart; Resource allocation suggestion	Notify PJM, PMO	Scheduler, Resource database
7	Critical Path Calculation	Schedule finalized	Auto-calculate critical path; Alert for delays	Notify PJM	Scheduler
8	Resource Planning & Optimization	Resource request	Auto-check availability; Reallocation suggestions	Notify HR for update	HR System
9	Procurement Setup	Procurement request	Auto-generate PR; Attach contract templates	Notify Procurement & Legal	Procurement System
10	Contract Management	Contract approval	Update contract repository; Assign contract number	Notify Legal	Contract Repository
11	Dashboard Setup	Project go-live	Auto-create monitoring dashboards; Auto-refresh	Notify PMO	Dashboard System
12	Field Data Collection	Site logging initiated	Real-time capture via mobile; Auto-sync data	Notify PJM	Dashboard
13	Portfolio Monitoring	Periodic update	Auto-KPI calculation; Summary email	Notify PMO, DIR	Reporting system
14	Cost Tracking	Transaction posting	Update budget report; Threshold alerts	Notify Finance, PJM	ERP System
15	Performance Monitoring	Project milestone reached	Generate scorecards; Escalate on low performance	Notify PMO, DIR	Performance Management
16	Risk Identification	Risk logged	Assign risk level; Recommend mitigations	Notify PJM, PMO	Risk Register
17	Mitigation Planning	Risk approval	Track mitigation actions; Send reminders	Notify PJM	Risk Register
18	Progress Reporting	Reporting cycle start	Auto-generate progress report	Notify stakeholders	Reporting
19	Reporting	Scheduled timeframe	Data aggregation; Distribute reports	Notify PMO, DIR	Reporting
20	Scenario Planning	Scenario initiated	Clone project plan; Highlight variances	Notify PJM	Planning Module
21	Benchmarking	Data completion	Compare metrics; Recommend improvements	Notify PMO	Analytics Engine
22	Closure	Project completion	Generate closure checklist; Archive data	Notify all stakeholders	Document Management
23	Knowledge Transfer	Project closure	Create knowledge entry; Invite for session	Notify project team	Knowledge Base

1.8. Automation Flow Map — Project Management

Process Stage	Automation Point	Automation Description	Trigger	Output
1. Project Initialization	EPS & WBS Auto-Setup Templates	Predefined templates for project structure	New project creation	Auto-created EPS and WBS hierarchy
	Configuration Auto-Sync	Sync configuration settings across projects	Initial setup complete	Standardized configuration applied
2. Portfolio Management	Auto-Portfolio Assignment	Assign projects to portfolios based on criteria	Project initiation	Portfolio view updated
	Automated Portfolio Dashboards	Real-time portfolio metrics update	Project data update	Updated dashboards
3. Planning & Estimation	Cost Estimation Automation	Auto-calculate costs based on WBS and resource rates	WBS created	Estimated budget populated
	Schedule Auto-Generation	Auto-generate project schedule based on task dependencies	WBS and resources defined	Draft project schedule
	Critical Path Alerts	Auto-identify and alert critical path tasks	Schedule changes	Alerts and notifications
4. Resource Management	Auto Resource Allocation	Suggest resource allocation based on availability	Task scheduling	Resource plan updated
	Resource Utilization Alerts	Notify over/under-utilization	Resource assignments	Alert notification
5. Cost & Performance Management	Cost Tracking Automation	Auto-track actual costs vs. baseline	Project expenses logged	Variance reports
	EVM (Earned Value Management) Automation	Auto-calculate project performance indices	Progress updates	SPI, CPI metrics
6. Procurement & Contracting	Supplier Auto-Match	Suggest suppliers based on procurement needs	Procurement request	Supplier shortlist
	Contract Expiry Alerts	Notify upcoming contract expiries	Contract dates	Expiry alerts
7. Execution & Monitoring	Auto Progress Updates	Auto-update project progress from field inputs	Field data collected	Progress dashboards
	Field Data Collection Automation	Auto-collect data via integrated forms/devices	Task execution	Data repository update
8. Risk Management	Risk Auto-Scoring	Calculate risk score based on inputs	Risk identification	Risk heatmap
	Auto-Mitigation Plan Suggestions	Propose mitigation plans from risk database	Risk scored	Suggested actions
9. Reporting & Analytics	Auto Report Generation	Scheduled or on-demand report generation	Data updates	Reports distributed
	Dashboard Auto-Refresh	Real-time dashboard updates	Any data changes	Live analytics
	Scenario Analysis Automation	Automated "what-if" analysis	Manual trigger	Scenario results
10. Closure & Knowledge Management	Auto Closure Checklists	Generate closure tasks checklist	Project completion	Closure plan
	Archive Automation	Auto-archive project data and documents	Project closure	Archived records
	Lessons Learned Auto-Collection	Auto-generate lessons learned draft from project logs	Project closure	Draft report for review

1.9. RACI Matrix — Project Management Process

For the recommended roles, we are proposing the below (adjustable to Wujha org):

- **PJM** = Project Manager
- **PMO** = Project Management Office
- **FIN** = Finance
- **PROC** = Procurement
- **ENG** = Engineering / Technical Team
- **SITE** = Site Supervisor / Field Team
- **QAQC** = Quality Assurance / Quality Control
- **IT** = IT / Automation Platform Support
- **DIR** = Project Director / Steering Committee
- **HR** = Human Resources (for resource planning)
- **LEGAL** = Legal Department (for contracts)

Notes:

- **PMO** plays a major Accountable role for governance, monitoring, and reporting.
- **PJM (Project Manager)** is Responsible for most operational tasks.
- **Finance** is naturally key in budgeting and cost tracking.
- **Procurement** and **Legal** have clear roles around contracting.
- **IT** is critical in system setup, dashboards, and field data capture automation.
- **Director / Steering Committee** is kept informed across, occasionally consulted for approvals.
- **HR** is lightly involved, especially around resource planning.

Activity	PJM	PMO	FIN	PROC	ENG	SITE	QAQC	IT	DIR	HR	LEGAL
1. Project Initialization	A	R	C	-	C	-	-	-	I	C	-
2. Configuration	R	A	C	-	C	-	-	R	I	I	-
3. Portfolio Creation	A	R	C	-	-	-	-	C	I	-	-
4. WBS Design	R	A	-	-	C	I	I	-	I	-	-
5. Cost Estimation & Allocation	R	C	A	C	C	-	-	-	I	-	-
6. Schedule Planning	R	A	I	-	C	I	-	-	I	-	-
7. Critical Path Calculation	R	A	I	-	C	I	-	-	I	-	-
8. Resource Planning & Optimization	R	C	C	-	C	-	-	-	I	A	-
9. Procurement Setup	I	C	I	A	C	-	-	-	I	-	C
10. Contract Management	C	I	C	R	I	-	-	-	I	-	A
11. Dashboard Setup	C	A	I	-	-	-	-	R	I	-	-
12. Field Data Collection	I	I	-	-	-	A	C	R	I	-	-
13. Portfolio Monitoring	I	A	C	-	C	I	-	C	I	-	-
14. Cost Tracking	C	R	A	-	-	I	-	-	I	-	-
15. Performance Monitoring	C	A	C	-	C	I	C	R	I	-	-
16. Risk Identification	R	A	I	-	C	I	C	-	I	-	-
17. Mitigation Planning	R	A	I	-	C	I	C	-	I	-	-
18. Progress Reporting	R	A	C	-	C	I	C	C	I	-	-
19. Reporting	R	A	C	-	C	I	C	C	I	-	-
20. Scenario Planning	R	A	C	-	C	I	-	C	I	-	-
21. Benchmarking	R	A	C	-	C	I	-	C	I	-	-
22. Closure	R	A	C	-	C	I	C	-	I	-	-
23. Knowledge Transfer	R	A	I	-	C	I	C	-	I	-	-

Legend:

- **R** = Responsible (Executes the task)
- **A** = Accountable (Owner of the task)
- **C** = Consulted (Needs to be consulted)
- **I** = Informed (Kept in the loop)

1.10. Attributes Table for Project Management System

#	Attribute Name	Description	Data Type	Mandatory (Y/N)	Source/Integration
1	Project ID	Unique identifier for each project	String	Y	Auto-generated
2	Project Name	Name/title of the project	String	Y	User Input
3	Project Description	Detailed description of the project	Text	Y	User Input
4	EPS Level	Enterprise Project Structure level assigned	String	Y	System / PMO
5	Portfolio ID	Identifier of the portfolio	String	Y	Auto-generated
6	WBS Code	Work Breakdown Structure code	String	Y	Auto-generated
7	WBS Description	Description of WBS element	Text	Y	User Input
8	Task ID	Task identifier in the schedule	String	Y	Auto-generated
9	Task Description	Detailed description of task	Text	Y	User Input
10	Task Dependency	Predecessor and successor tasks	String	Y	System Calculated
11	Critical Path Flag	Indicator for critical tasks	Boolean	Y	System Calculated
12	Start Date	Planned start date of task/project	Date	Y	User Input
13	End Date	Planned end date of task/project	Date	Y	User Input
14	Actual Start Date	Actual start date (field updates)	Date	N	Field Data Collection
15	Actual End Date	Actual completion date	Date	N	Field Data Collection
16	Budget Amount	Total budget allocated	Currency	Y	Finance System
17	Cost Element	Budget distribution per WBS element	Currency	Y	Finance System
18	Actual Cost	Actual spent amount	Currency	N	ERP Integration
19	Resource ID	Unique identifier of resource	String	Y	HR System
20	Resource Role	Role of the assigned resource	String	Y	HR System
21	Resource Availability	Availability status of resource	String	Y	HR System
22	Procurement ID	Procurement reference number	String	Y	Procurement System
23	Contract ID	Contract reference number	String	Y	Contract Management
24	Risk ID	Unique identifier of risk	String	Y	Risk Register
25	Risk Level	Severity level (High/Medium/Low)	String	Y	Risk Assessment
26	Mitigation Plan	Description of mitigation action	Text	N	Risk Register
27	KPI Name	Key performance indicator name	String	Y	Dashboard System
28	KPI Value	Value of KPI	Decimal	Y	Dashboard System
29	Scenario ID	"What-if" scenario identifier	String	Y	System Generated
30	Benchmark Reference	Reference to benchmark dataset	String	N	Analytics System
31	Document Link	Link to project document repository	URL	Y	Document Management
32	Project Status	Current status (Planning, Execution, Closed)	String	Y	System Calculated
33	Progress %	Percentage of completion	Percentage	Y	System Calculated
34	Report ID	Report reference number	String	Y	Reporting System
35	Notification Type	Type of notification (Email, System, SMS)	String	Y	Notification Engine
36	Stakeholder List	List of stakeholders	String	Y	User Input / System
37	Lessons Learned	Summary of post-project insights	Text	N	User Input
38	Closure Checklist	List of completed closure activities	Text/List	Y	System Generated

1.11. Visual Data Flow mapped to the above Attributes:

Step	Input Data	Process	Output Data	Destination / Next Step
1	Project Request, Project Charter	Project Initialization (EPS/WBS Setup)	Project Structure Data (EPS/WBS IDs, names, levels)	Configuration
2	Project Structure Data	Configuration (Align hierarchy)	Configured Project Hierarchy	Portfolio Creation
3	Configured Project Hierarchy	Portfolio Creation	Project Portfolio Data	Portfolio Monitoring
4	Project Portfolio Data, Progress Updates	Portfolio Monitoring	Portfolio Performance Data	Dashboard & Reporting
5	Project Scope, Deliverables	WBS Design	Detailed WBS Elements	Cost Estimation & Schedule Planning
6	WBS, Historical Costs, Estimates	Cost Estimation & Allocation	Budget Allocation	Schedule Planning
7	WBS, Budget Allocation	Schedule Planning	Project Schedule (tasks, durations, dependencies)	Critical Path Calculation
8	Project Schedule	Critical Path Calculation	Critical Path Data	Resource Planning
9	Project Schedule, Resource Pool	Resource Planning	Initial Resource Assignments	Resource Optimization
10	Resource Assignments, Availability Data	Resource Optimization	Optimized Resource Plan	Cost Tracking
11	Schedule, Budget, Resource Plan	Cost Tracking	Time-Phased Cost Data	Performance Monitoring
12	Cost Data, Progress Updates	Performance Monitoring (EVM)	Performance Variance Reports	Dashboard, Reports
13	Procurement Plan, Vendor List	Procurement Setup	Procurement Requirements	Contract Management
14	Supplier Data, Procurement Requirements	Contract Management	Contract Documents, Supplier Data	Procurement & Progress Reporting
15	Task Updates, Site Reports	Field Data Collection	Field Progress Data	Progress Reporting
16	Progress Data, Schedule Data	Progress Reporting	Progress Reports	Dashboard & Risk Identification
17	Project Data, Market Insights	Risk Identification	Risk Register	Mitigation Planning
18	Risk Register, Risk Analysis	Mitigation Planning	Mitigation Strategies	Dashboard, Reports
19	Project KPIs, Monitoring Data	Dashboard Setup	Configured Dashboard	Portfolio Monitoring
20	Reports, Portfolio Data	Reporting	Distributed Reports	Stakeholders
21	Project Deliverables, Contracts	Closure	Closure Checklist, Sign-off	Knowledge Transfer
22	Closure Data, Lessons	Knowledge Transfer	Lessons Learned Repository	Scenario Planning
23	Project Data, Historical Data	Scenario Planning	Scenario Plans	Benchmarking
24	Scenario Plans, Industry Benchmarks	Benchmarking	Benchmark Reports	Portfolio Monitoring, Planning

• **Approvals:**

- Approver 1

- Approver 2
